



DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Rajiv Gandhi University of Knowledge Technologies – Nuzvid

Nuzvid, Eluru, Andhra Pradesh – 521202.

Interactive Visual Aid

A Project Progress Report

Submitted in partial fulfillment for the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Submitted by

PASALA BALAJI (N190868)

Under the Esteem Guidance of

Mr. Gummadidala Jaya Krishna



DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Rajiv Gandhi University of Knowledge Technologies – Nuzvid

Nuzvid, Eluru, Andhra Pradesh – 521202.

CERTIFICATE OF COMPLETION

This is to certify that the work entitled, “**Interactive Visual Aid**” is the bonafied work of **PASALA BALAJI (ID : N190868)** carried out under my guidance and supervision for 4th year project of **Bachelor of Technology** in the department of Computer Science and Engineering under RGUKT IIIT Nuzvid. This work is done during the academic session December 2024 to April 2025, under our guidance.

Mr. Gummadidala Jaya Krishna

Assistant professor,
Department of CSE,
RGUKT Nuzvid

Mrs. Samineni Bhavani

Assistant Professor,
Head of the Department,
Department of CSE,
RGUKT Nuzvid.



DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Rajiv Gandhi University of Knowledge Technologies – Nuzvid

Nuzvid, Eluru, Andhra Pradesh – 521202.

CERTIFICATE OF EXAMINATION

This is to certify that the work entitled, “**Interactive Visual Aid**” Completion is the Bonafide work of **PASALA BALAJI (ID : N190868)**, and here by accord our approval of it as a study carried out and presented in a manner required for its acceptance in 4th year of **Bachelor of Technology** for which it has been submitted. This approval does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn, as a recorded in this thesis. It only signifies the acceptance of this thesis for the purpose for which it has been submitted.

Mr. Gummadidala Jaya Krishna

Assistant Professor,

Department of CSE,

RGUKT-NUZVID

Mr. K. Sravan Kumar

Assistant Professor,

Department of CSE,

RGUKT- NUZVID



DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Rajiv Gandhi University of Knowledge Technologies – Nuzvid

Nuzvid, Eluru, Andhra Pradesh – 521202.

DECLARATION

I “**PASALA BALAJI (ID : N190868)**”, hereby declare that the project report entitled “**Interactive Visual Aid**” done by us under the guidance of Mr. Gummadidala Jaya Krishna, Assistant Professor is submitted for the partial fulfillment for the award of degree of Bachelor of Technology in Computer Science and Engineering during the academic session December 2024 - April 2025 at RGUKT-Nuzvid.

I also declare that this project is a result of my own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

Date: 25-04-2025,

Place: Nuzvid.

PASALA BALAJI (ID : N190868)

ACKNOWLEDGEMENT

We would like to express our profound gratitude and deep regards to our guide **Mr. Gummadidala Jaya Krishna** for his exemplary guidance, monitoring and constant encouragement to us throughout the B.Tech course. We shall always cherish the time spent with him during the course of this work due to the invaluable knowledge gained in the field of reliability engineering.

We are extremely grateful for the confidence bestowed in us and entrusting our project entitled **“Interactive Visual Aid”**

We express gratitude to **Mrs. Samineni Bhavani** (HOD of CSE) and other faculty members for being source of inspiration and constant encouragement which helped us in completing the project successfully.

Finally, yet importantly, we would like to express our heartfelt thanks to our beloved God and parents for their blessings, our friends for their help and wishes for the successful completion of this project.

ABSTRACT

Interactive Visual Aid (IVA) is an innovative Progressive Web Application (PWA) designed to ensure accessibility to learning and reference materials irrespective of network availability. It allows users to access essential digital content such as PDF documents, instructional videos, presentations, and images both online and offline. This is achieved through the use of modern browser technologies like the Cache API and Indexed DB, which enable the application to locally store content for future use. Users can pre-download resources while online and continue using them offline, providing a seamless and efficient experience, especially in remote or bandwidth-constrained areas. IVA significantly contributes to bridging the digital divide by enhancing educational and informational access in low-connectivity environments.

TABLE OF CONTENTS

ABSTRACT	6
CHAPTER 1	8
INTRODUCTION	8
1.1 Overview of IVA	8
1.2 Importance of Offline-First Design	8
1.3 Features and Capabilities	8
1.4 Real-World Applications	8
CHAPTER 2	9
REQUIREMENTS AND ANALYSIS	9
2.1 Hardware Requirements	9
2.2 Software Stack	9
2.3 Functional Requirements	10
2.4 Non-Functional Requirements	10
CHAPTER 3	11
PROPOSED MODEL AND FLOW OF THE PROJECT	11
3.1 System Workflow	11
3.2 Advantages of the Proposed Model	12
CHAPTER 4	13
IMPLEMENTATION AND OUTPUTS	13
4.1 Implementation	13
4.2 Workflow	14
4.3 Outputs	15
CHAPTER 5	16
FUTURE WORKS AND CONCLUSION	16
5.1 Future Scope	16
5.2 Conclusion	18
CHAPTER 6	19
REFERENCES	19

CHAPTER 1

INTRODUCTION

In today's increasingly digital world, uninterrupted access to web-based content is often assumed, yet many regions and users still suffer from unreliable or non-existent internet connectivity. The motivation behind the development of IVA stems from the need to support such users—students in rural areas, professionals in the field, and travelers—by offering an offline-first experience. By allowing users to download and cache essential content when they have network access, IVA ensures that their learning and reference experience is not hindered by connectivity issues. Unlike traditional web apps, IVA functions effectively as a standalone application on both mobile and desktop platforms, thanks to its PWA architecture, providing app-like capabilities such as installability, offline usage, and background syncing.

1.1 Overview of IVA

IVA is a Progressive Web Application (PWA) designed to deliver content such as PDFs, images, presentations, and videos both online and offline. It combines the accessibility of the web with the reliability of native applications, allowing users to continue using the platform even without an internet connection.

1.2 Importance of Offline-First Design

The core principle of IVA is its offline-first architecture. It allows users to download and cache essential content while online and access it seamlessly during offline periods. This ensures learning continuity and operational efficiency for users in bandwidth-constrained environments.

1.3 Features and Capabilities

IVA offers key features including:

- Installability on mobile and desktop
- Offline access using Cache API and Indexed DB
- Background syncing when connectivity returns
- Responsive user interface and real-time interaction

1.4 Real-World Applications

IVA's use cases are extensive and include:

- Students attending classes in areas with intermittent internet
- Field workers needing reference material on the go
- Organizations seeking cost-effective content distribution in remote area

CHAPTER 2

REQUIREMENTS AND ANALYSIS

2.1 Hardware Requirements

The IVA system is designed to be lightweight and accessible, ensuring it runs efficiently on commonly available hardware without the need for high-end specifications. The following are the recommended minimum hardware requirements:

- **Processor:** Dual-core or quad-core CPU (1.8 GHz or higher)
- **RAM:** Minimum 2 GB (4 GB recommended for smoother performance)
- **Storage:** 500 MB to 1 GB free space for cached resources (depending on user content)
- **Display:** Any modern browser-enabled screen (mobile, tablet, laptop)
- **Network:** Internet required only for initial setup and content download (optional afterward)

This ensures compatibility with low-cost devices, enabling use in developing regions or educational institutions with basic infrastructure.

2.2 Software Stack

IVA uses a modular software architecture, integrating several key components for both backend and frontend functionality.

Backend Technologies:

- **Django:** A high-level Python web framework used for managing HTTP requests, content delivery, API endpoints, and session management. Chosen for its scalability, built-in security features, and rapid development capability.
- **PostgreSQL:** An open-source relational database system used to store user profiles, access logs, content metadata, and system configurations.

Frontend Technologies:

- **JavaScript (Vanilla + Progressive Enhancements):** Enables dynamic UI rendering, offline state detection, and user event handling.
- **Service Workers:** Intercept network requests and manage caching for offline access.
- **Indexed DB:** A NoSQL browser database that allows structured storage of large objects like PDFs and videos.
- **Cache API:** Stores static files such as HTML, CSS, JS, and images to support full offline rendering.

Progressive Web Application (PWA) Tools:

- Web App Manifest (for install ability)
- Responsive design using HTML5 and CSS3
- Browser-native APIs for device storage and background sync

2.3 Functional Requirements

IVA must fulfill the following core functionalities:

- **Offline Resource Access:** Users should be able to access previously downloaded PDFs, videos, and images when disconnected.
- **Content Caching:** All static and media content must be cacheable using browser APIs.
- **Background Syncing:** User activity while offline should be logged and synced to the server upon reconnection.
- **User Interface Responsiveness:** Application must adapt to different screen sizes and network conditions.
- **Content Filtering:** Search and filter capabilities should be available to find specific content types.
- **Content Update Notification:** Users should be notified if newer versions of downloaded resources are available when they reconnect.

2.4 Non-Functional Requirements

These define the overall quality and system characteristics:

- **Performance:** Application should load within 3 seconds when offline using cached assets.
- **Scalability:** Must support hundreds of users concurrently in a cloud-hosted environment.
- **Security:**
 - User data must be encrypted when stored in Indexed DB .
 - Authentication via secure tokens/session cookies for content access.
- **Maintainability:**
 - Modular codebase for easy updates and upgrades.
 - Configuration-driven deployment for fast scaling.
- **Compatibility:**
 - Support for all modern browsers (Chrome, Edge, Firefox, Safari)
 - Device-independent operation (runs on mobile, tablet, and desktop)
- **Accessibility:** UI should be compliant with WCAG 2.1 standards to support users with disabilities.

CHAPTER 3

PROPOSED MODEL AND FLOW OF THE PROJECT

3.1 System Workflow

The IVA application follows a well-defined architecture to manage both online and offline user experiences. This section describes each component and step of the flow in detail.

Step 1: User Initialization

When a user accesses the IVA application via a web browser, the system checks for network connectivity. If internet access is available, the server delivers the latest version of the web interface and associated resources. The application can also be installed as a PWA on the user's device, mimicking the feel of a native app.

Step 2: Resource Preloading

Once online, users can select which learning assets they want to download for offline use. These may include:

- PDF documents (notes, textbooks, guides)
- Video lectures
- Image-based content (infographics, diagrams)
- Presentation slides

The user-initiated download triggers the browser to store these resources locally using Cache API and Indexed DB.

Step 3: Caching and Storage

To facilitate offline access:

- **Cache API** stores static assets like HTML, CSS, JavaScript files, and images. This ensures the app layout and design are always available.
- **Indexed DB** is used to store complex and dynamic assets such as PDFs, video files, user logs, and metadata. Indexed DB allows structured data storage, making it highly suitable for offline applications.

This dual-storage approach guarantees that both the interface and the actual educational content are accessible without an internet connection.

Step 4: Offline Usage

Once the user goes offline, the application detects the network loss and switches to offline mode. Since both the interface and selected content are stored locally:

- Users can continue reading documents or watching downloaded videos.
- The application remains fully functional except for server-dependent features (like user authentication or live updates).
- All user interactions are recorded in the background and stored in a local queue.

Step 5: Data Synchronization

When connectivity is restored:

- The app automatically detects the change in network status.
- Locally stored activity logs, such as which resources were accessed and for how long, are synced to the backend database.
- The system checks for any updated content on the server and prompts the user to refresh outdated materials.

This process ensures that both the user experience and system records remain consistent across sessions and devices.

3.2 Advantages of the Proposed Model

- **Resilience in Connectivity-Challenged Areas:** Users are no longer dependent on real-time internet access to continue learning or consuming content.
- **Reduced Bandwidth Consumption:** Since assets are stored locally, repeat downloads are avoided, reducing server load and data usage.
- **Improved Performance:** The application loads quickly and runs smoothly even without a network, enhancing the overall user experience.
- **Data Consistency:** The system ensures that once the connection is restored, all relevant data is synced automatically, maintaining consistency across online and offline modes.

CHAPTER 4

IMPLEMENTATION AND OUTPUTS

4.1 Implementation

The development of IVA followed a modular and scalable approach, ensuring that the system remains maintainable, extensible, and efficient across various use cases.

Backend Implementation (Django Framework)

The backend is developed using Django, a Python-based framework known for its robustness and built-in features. Key responsibilities of the backend include:

- **API Development:** RESTful APIs were designed to handle requests for content metadata, user authentication, session management, and analytics.
- **Database Management:** PostgreSQL was used to manage structured data such as user profiles, content usage logs, and resource indexing.
- **Security:** Django's in-built middleware and session management were used to secure content access and handle permissions efficiently.

Frontend Implementation (JavaScript and PWA Architecture)

The frontend was designed to be responsive and intelligent, using:

- **JavaScript:** For DOM manipulation, state management, and UI event handling. It dynamically updates the interface based on the user's connectivity status.
- **Service Workers:** These enable the offline-first functionality by intercepting HTTP requests and serving cached content instead of reaching out to the server.
- **Web App Manifest:** Ensures the PWA is installable and provides metadata such as app icons, startup behavior, and theme colors.
- **Indexed DB and Cache API:** Serve as the local storage layers. Cache API handles lightweight static files (e.g., stylesheets, scripts), while Indexed DB is used to persist rich content like video lectures, documents, and user session data.

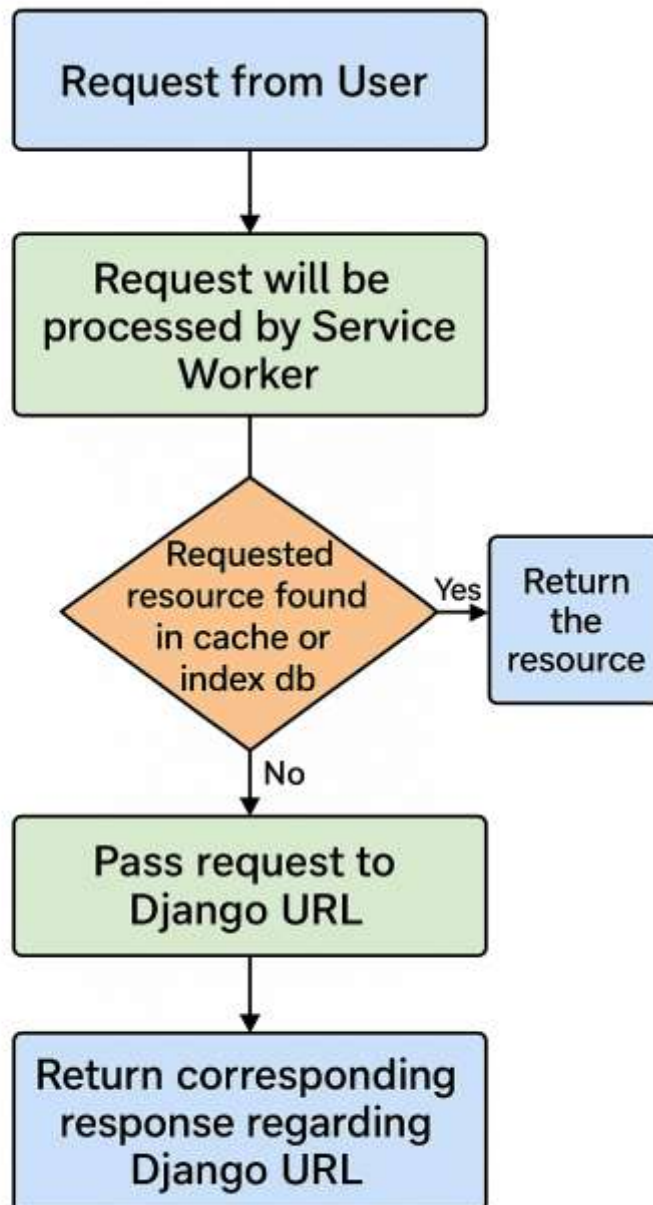
Offline Interaction and Synchronization

A critical part of implementation was managing how the system behaves offline:

- All user interactions (e.g., opening a PDF, watching a video) are logged in a local queue.
- When the network becomes available again, a background synchronization routine is triggered to send these logs to the backend for permanent storage.
- If content updates are detected on the server, the application notifies the user and optionally updates the cached version.

This dual-mode operation—online and offline—ensures continuity and a smooth user experience regardless of connectivity status.

4.2 Workflow



4.2 Outputs

The system was rigorously tested to ensure that the intended functionality is achieved across various environments, such as low-bandwidth zones, mobile devices, and long offline durations. The following are the observed outputs:

Smooth Rendering of Cached Assets

Once a user downloads content such as PDFs or videos, these files are cached locally. When the user revisits the application offline:

- Static files (HTML, CSS, JS) are served instantly from the browser cache.
- Media content loads without buffering since it resides in Indexed DB. This results in a fast, fluid user experience even with no network connection.

Seamless Offline User Interaction

All core interactions—navigating menus, viewing resources, switching between sections—remain fully functional offline. The frontend UI detects network status changes in real-time and adapts automatically:

- Offline indicators are shown.
- Online-only features are temporarily disabled or queued.
- Downloads requested while offline are deferred and processed once online.

Background Sync and Real-Time Asset Updates

A background task manager checks periodically for:

- New versions of previously downloaded content.
- Pending logs or data that need to be synced with the server. If updates are available, the user is notified with options to update their offline copy.

Cross-Device Compatibility

IVA was tested across:

- Browsers: Chrome, Firefox, Edge, Safari
- Platforms: Android, iOS, Windows, macOS
- Screen Sizes: Phones, Tablets, Desktops

Responsive design ensures optimal display and usability across all device types. No installation is required, though users can optionally install it as a standalone app on their home screen (via PWA capabilities).

CHAPTER 5

FUTURE WORKS

5.1 Future Scope

As IVA continues to evolve, several enhancements are envisioned to improve its capabilities, broaden its application scope, and offer a more personalized, intelligent experience for users. These proposed future developments include:

A. Offline User Input and Data Capture

Currently, IVA allows users to access content offline but does not yet support data submission during offline sessions. A major enhancement will be enabling users to:

- Fill out feedback forms
- Submit quizzes or assignments
- Upload images, documents, or text notes

These inputs will be **stored locally using Indexed DB** and **synchronized with the backend** once the device reconnects to the internet. This would significantly increase interactivity and usefulness, especially for students in remote learning setups or field workers collecting data in offline conditions.

B. Intelligent Analytics and Usage Insights

IVA aims to implement advanced analytics modules that analyze:

- Which content is accessed most frequently
- Duration of engagement with videos or documents
- Device and platform usage patterns
- Offline vs. online behavior trends

Such data can help institutions and content creators improve the quality of their materials. These insights could be visualized via interactive dashboards on the admin panel, assisting educators or administrators in making data-driven decisions.

C. Personalized Content Recommendations

Leveraging access logs and historical user behavior, IVA could offer **context-aware recommendations**, such as:

- “You might also like...” resources based on subject matter
- Prioritized updates for downloaded resources
- Suggested videos or PDFs aligned with previous learning activity

This will be driven by lightweight machine learning algorithms running either on-device or server-side, respecting privacy while delivering a smarter experience.

D. Integration with Learning Management Systems (LMS)

To increase IVA's adoption in formal educational environments, future updates will support:

- **LMS integration protocols** such as LTI (Learning Tools Interoperability)
- Synchronization of grades, content, and user credentials with platforms like Moodle, Google Classroom, or Canvas
- Role-based access for teachers and students, allowing differentiated interaction and monitoring

Such integration will make IVA a plug-and-play solution for schools, colleges, and training institutions seeking robust offline access functionality.

E. Accessibility and Multi-language Support

Planned updates include:

- Support for screen readers and keyboard navigation (WCAG compliance)
- Multi-language content filtering and translation capabilities
- Font resizing and high-contrast modes for users with visual impairments

These features will broaden IVA's reach to diverse populations and align it with inclusive design principles.

5.2 Conclusion

IVA (Interactive Visual Aid) is a transformative solution built to tackle one of the most persistent barriers in digital education and content access: **network dependence**. By enabling users to download, store, and interact with digital resources offline, it ensures uninterrupted learning and information flow regardless of external conditions.

The system's architecture—powered by Django, JavaScript, Cache API, and Indexed DB—provides a seamless experience that mimics native applications while being accessible through standard web browsers. From smooth offline rendering to intelligent syncing and future-forward adaptability, IVA lays the groundwork for **inclusive, scalable, and resilient web solutions**.

In a world increasingly reliant on digital infrastructure, IVA positions itself as a powerful tool for institutions, learners, and communities seeking **equity in access, efficiency in performance, and continuity in learning**. With continued development and strategic integration, IVA holds the promise of becoming an essential component of modern digital ecosystems, particularly in **education, field research, remote work, and disaster-resilient communication**.

CHAPTER 6

REFERENCES

- [1] Mozilla, "Indexed DB API," *MDN Web Docs*, [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/IndexedDB_API [Accessed: May 1, 2025].
- [2] Mozilla, "Cache," *MDN Web Docs*, [Online]. Available: <https://developer.mozilla.org/en-US/docs/Web/API/Cache> [Accessed: May 1, 2025].
- [3] Django Software Foundation, "The Web framework for perfectionists with deadlines," *Django*, [Online]. Available: <https://www.djangoproject.com/> [Accessed: May 1, 2025].
- [4] PostgreSQL Global Development Group, "PostgreSQL: The world's most advanced open source database," *PostgreSQL*, [Online]. Available: <https://www.postgresql.org/> [Accessed: May 1, 2025].